

Notes: Avramov, Ge, Li, and Linton (JFE, 2026)

Dual Peer Effects and Cross-Stock Predictability

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1 What the paper is doing

The paper builds an *economically structured* cross-stock return predictor called the *Peer Index* (PI). Instead of predicting a stock’s return only from its own characteristics, PI uses a peer network (industry peers, shared-analyst peers, or customer–supplier peers) to map *peer information* into the stock’s expected return. Conceptually, PI decomposes the information that matters for stock i into two channels: (i) **peer strength** (how strong i ’s peers are on average) and (ii) **within-peer position** (how unusual i is relative to its peers). Empirically, PI predicts both stock returns and earnings surprises over short and long horizons. Importantly, even powerful machine-learning (ML) expected-return proxies that use *only own-stock characteristics* do not subsume PI, which supports the interpretation that PI captures *genuine cross-stock information* rather than merely a better nonlinear transformation of a stock’s own signals. Finally, the paper provides dynamic evidence (lag-augmented local projections) consistent with *slow diffusion* of peer information into prices (positive PI shocks predict gradually decaying future returns without reversal).

2 Notation and objects

- Time t is monthly (the main return-prediction setting).
- There are N_t stocks at time t , indexed by i .
- $R_{i,t+1}$: (excess) return of stock i from t to $t + 1$.
- $x_{i,t} \in \mathbb{R}^S$: a vector of S firm-level characteristics for stock i at time t . The paper uses a large set (Appendix Table 12).
- A peer network determines, for each stock i , a peer set g_i (e.g., SIC-industry peers, shared-analyst peers, or customer–supplier peers).

3 Core framework: dual peer effects

3.1 Baseline return prediction

A general return-prediction problem can be written as:

$$R_{i,t+1} = h(x_{i,t}) + e_{i,t+1}, \quad (1)$$

where $h(\cdot)$ is the (possibly nonlinear) mapping from characteristics to expected returns.

A standard linear single-characteristic specialization is

$$R_{i,t+1} = x_{s,i,t} \beta + e_{i,t+1}, \quad s \in \{1, \dots, S\}. \quad (2)$$

3.2 Peer average and deviation-from-peers

Given a peer set g_i , define for each characteristic s :

$$z_{s,i,t} \equiv \frac{1}{|g_i|} \sum_{j \in g_i} x_{s,j,t} \quad (\text{peer average / peer strength}), \quad (3)$$

$$\tilde{x}_{s,i,t} \equiv x_{s,i,t} - z_{s,i,t} \quad (\text{deviation from peers / within-peer position}).$$

If one simply substitutes $x_{s,i,t} = z_{s,i,t} + \tilde{x}_{s,i,t}$ into Eq. (2), one obtains

$$R_{i,t+1} = z_{s,i,t} \beta + \tilde{x}_{s,i,t} \beta + e_{i,t+1}, \quad (4)$$

which mechanically imposes that the peer-average and deviation components enter with the *same* slope β .

3.3 Dual peer effects model

The key conceptual move of the paper is to *relax* that restriction and allow peer averages and deviations to have different pricing/predictive roles:

$$R_{i,t+1} = z'_{i,t} \beta_z + \tilde{x}'_{i,t} \beta_x + e_{i,t+1}, \quad (5)$$

where $z_{i,t} \in \mathbb{R}^S$ stacks $\{z_{s,i,t}\}_{s=1}^S$ and $\tilde{x}_{i,t} \in \mathbb{R}^S$ stacks $\{\tilde{x}_{s,i,t}\}_{s=1}^S$.

Economic interpretation (the “two channels”).

- **Peer-average channel ($z_{i,t}$):** information embedded in the peer group. In an industry network, $z_{i,t}$ captures the *state of the industry* (e.g., peer profitability, investment, liquidity, etc.). This is naturally related to factor exposures implied by characteristics.
- **Deviation-from-peers channel ($\tilde{x}_{i,t}$):** information in how a firm differs from its peers. This captures *relative positioning* (e.g., “cheap vs expensive relative to industry”, “high vs low investment relative to industry”).

Both channels can forecast returns, and they need not do so with the same sign or magnitude.

4 Peer networks as a position matrix

4.1 Industry linkage matrix

For the baseline peer definition, the paper uses SIC-based industry peers. Let L^{ind} be an $N \times N$ linkage matrix with entries

$$\left(L^{\text{ind}}\right)_{ij} \equiv \mathbb{I}\{i \text{ and } j \text{ in same industry}\} \cdot \frac{1}{|g_i|}. \quad (6)$$

This is a row-normalized adjacency matrix, so each row averages over i ’s peers.

4.2 Matrix expression for peer averages

Let $x_{s,t} \in \mathbb{R}^N$ be the cross-section of characteristic s at time t . Then for a general position matrix L ,

$$z_{s,i,t} = x'_{s,t} L_i, \quad \tilde{x}_{s,i,t} = x_{s,i,t} - z_{s,i,t}, \quad (7)$$

where L_i is the i th column (or equivalently, L_i extracts the appropriate weighted average for i).

5 How PI is constructed in practice

5.1 Ridge estimation of the dual peer effects mapping

Operationally, PI is the *fitted value* from a high-dimensional version of Eq. (5) using all S characteristics:

$$R_{i,t+1} \approx z'_{i,t} \beta_z + \tilde{x}'_{i,t} \beta_x.$$

Because S is large and the regressors are correlated, the paper uses ridge regularization.

A generic ridge objective for estimating (β_z, β_x) is:

$$\min_{\beta_z, \beta_x} \sum_{(i,t) \in \mathcal{W}} (R_{i,t+1} - z'_{i,t} \beta_z - \tilde{x}'_{i,t} \beta_x)^2 + \lambda (\|\beta_z\|_2^2 + \|\beta_x\|_2^2), \quad (1)$$

where $\mathcal{W}$ denotes the estimation window and $\lambda > 0$ is a tuning parameter.

PI signal. Given estimated coefficients, define the raw PI forecast:

$$\hat{R}_{i,t+1}^{PI} \equiv z'_{i,t} \hat{\beta}_z + \tilde{x}'_{i,t} \hat{\beta}_x.$$

The paper then standardizes / ranks PI each month so that the signal is comparable across time and interpretable in cross-sectional regressions and portfolio sorts.

5.2 Why PI is not “just ML”

A key comparison throughout the paper is between:

- ML predictors that only use *own-stock* characteristics $x_{i,t}$, and
- PI, which uses peer averages and peer deviations $(z_{i,t}, \tilde{x}_{i,t})$.

If PI remains predictive after controlling for ML expected-return proxies, that is interpreted as evidence that PI captures *cross-stock* information content that cannot be recovered from $x_{i,t}$ alone.

6 Connection to principal portfolios and attention-induced pricing

6.1 Attention-induced pricing model (AIPM)

The paper notes an AIPM-style portfolio-weight mapping:

$$w_{i,t}^{AIPM} = \lambda' X'_t A_i, \quad (8)$$

where X_t stacks characteristics, A is an “attention” (or position) matrix, and A_i is the relevant column.

With A as the position matrix, one can write

$$z_{s,i,t} = \sum_{j=1}^N A_{ij} x_{s,j,t} = A'_i x_{s,t}, \quad z_{i,t} = X'_t A_i. \quad (9)$$

If $\lambda = \beta_z$, then

$$w_{i,t}^{AIPM} = \beta'_z z_{i,t}. \quad (10)$$

This highlights that the peer-average leg can be viewed as an economically motivated “position” in characteristic space.

6.2 Bayesian shrinkage toward industry linkages

Rather than fully learning A from scratch, a natural disciplined approach is to shrink toward the industry linkage matrix:

$$A_{ij} \sim \mathcal{N}\left(\left(L^{\text{ind}}\right)_{ij}, \tau^2\right). \quad (11)$$

This makes the industry network a prior mean and allows data-driven deviations.

7 Empirical designs and evaluation metrics

7.1 Out-of-sample performance: R^2 and prediction-error comparisons

Out-of-sample performance for model m is measured by

$$R_{OOS}^2 = 1 - \frac{\sum_t \sum_i \left(R_{i,t+1} - \hat{R}_{i,t+1|t}^m\right)^2}{\sum_t \sum_i R_{i,t+1}^2}, \quad (12)$$

with returns de-meaned appropriately (as in the paper).

To compare models by their mean squared prediction error (MSPE), the paper uses

$$d_t^{m,m'} \equiv \sum_i \left(R_{i,t+1} - \hat{R}_{i,t+1|t}^m\right)^2 - \sum_i \left(R_{i,t+1} - \hat{R}_{i,t+1|t}^{m'}\right)^2, \quad (13)$$

and tests whether the average $d_t^{m,m'}$ differs from zero (Diebold–Mariano logic).

7.2 Peer Index factor (PIF) used for factor spanning

To study whether PI can be spanned by standard factors, the paper forms a *peer index factor* (PIF) by averaging PI long–short spreads in big and small stocks:

$$PIF_t \equiv \frac{1}{2}PIF_t^{big} + \frac{1}{2}PIF_t^{small}. \quad (14)$$

This construction is designed to mitigate concerns that predictability is mechanically driven by microcaps while retaining broad coverage.

7.3 Dynamics: lag-augmented local projections (LA-LP)

To study the dynamic return response to a PI shock, the paper estimates:

$$R_{i,t+h} = \alpha^{(h)} + \beta^{(h)} PI_{i,t} + \sum_{j=1}^{12} \phi_j^{(h)} R_{i,t-j} + \sum_{j=1}^{12} \psi_j^{(h)} PI_{i,t-j} + \nu_{i,t}^{(h)}, \quad (15)$$

and plots $\beta^{(h)}$ across horizons h .

Interpretation. If PI captures *underreaction* to peer information, then a positive PI innovation should predict *positive future returns* that decay *gradually* and should not reverse strongly.

7.4 Fundamentals (earnings surprises) regressions

To connect PI to fundamentals, the paper studies whether PI predicts standardized unexpected earnings (SUE). At the industry level:

$$SUE_{ind,i,t,t+1} = \alpha + \beta PI_{i,t} + \gamma ML_{i,t} + \Gamma' W_{i,t} + \varepsilon_{i,t+1}, \quad (16)$$

and similarly for industry-level returns.

At the firm level, for horizon h :

$$SUE_{i,t+h} = \alpha + \beta PI_{i,t} + \gamma ML_{i,t} + \Gamma' W_{i,t} + \varepsilon_{i,t+h}. \quad (17)$$

8 Main results: interpreting each table and figure

8.1 Table 1: Fama–MacBeth cross-sectional regressions

Object: monthly value-weighted Fama–MacBeth regressions of $R_{i,t+1}$ on PI and standard predictors.

- **Panel A (basic controls):** PI enters with a large, positive, and statistically significant slope after controlling for short-term reversal and momentum, and after controlling for either industry momentum or a lead–lag term. Interpretation: the signal is not a relabeling of standard momentum/reversal or simple industry momentum.
- **Panel B (robustness):** PI remains significant when controlling for ML expected-return proxies. Interpretation: own-characteristic ML does not subsume the peer-based signal; this supports the “genuine cross-stock information” view.
- **Panel C (window settings):** PI remains predictive under alternative estimation-window choices. Interpretation: the result is not fragile to a particular training-window design.

- **Panel D (long-horizon predictability):** PI predicts multi-month cumulative returns. Interpretation: predictability is not purely one-month reversal; it persists.
- **Panel E (market state):** PI predicts in both “high” and “low” states. Interpretation: PI is not only a crisis artifact.
- **Panel F (nonlinearity):** a nonlinear version of PI has additional explanatory power. Interpretation: the PI mapping can benefit from mild nonlinearity, but the core peer logic remains.

Table 1
Value-weighted Fama-MacBeth cross-sectional regressions.

Panel A: Basic controls					
Dependent variable	Observations	PI	MOM1	MOM12	INDMOM
$R_{i,t}$	2,088,332		-0.68*** (-3.19)	0.83** (2.19)	0.51*** (3.15)
$R_{i,t+1}$	2,088,332	0.99*** (5.90)	0.58 (1.15)	0.58 (1.44)	0.06 (0.36)
$R_{i,t+12}$	2,088,332		-0.78*** (-3.78)	0.86** (2.56)	0.60*** (3.56)
$R_{i,t+120}$	2,088,332	0.85*** (5.71)	-0.41** (-2.00)	0.60 (1.60)	0.33* (1.80)
Panel B: Robustness					
Dependent variable	Observations	PI			
$R_{i,t}$	With ML control (GBRT)	2,088,332	0.81*** (4.20)		
$R_{i,t+1}$	With ML control (RF)	2,088,332	0.78*** (4.17)		
$R_{i,t+12}$	With ML control (NN)	2,088,332	0.81*** (3.15)		
$R_{i,t+120}$	With all controls	2,088,332	0.50*** (3.87)		
$R_{i,t}$	Remove microcaps	1,043,044	0.92*** (4.61)		
$R_{i,t+120}$	Post 2003 Sample	942,090	1.96*** (7.47)		
Panel C: Window settings					
Dependent variable	Window size	Observations	PI	PI post 2003	
$R_{i,t}$	Rolling 12	2,088,332 (942,090)	2.57*** (8.68)	1.46*** (5.85)	
$R_{i,t+1}$	Rolling 60	2,088,332 (942,090)	2.81*** (9.64)	1.35*** (5.72)	
$R_{i,t+12}$	Rolling 120	2,088,332 (942,090)	1.50*** (10.21)	1.57*** (6.69)	
$R_{i,t+120}$	Rolling 360	2,088,332 (942,090)	3.26*** (11.60)	1.96*** (7.31)	
$R_{i,t}$	Expanding	2,088,332 (942,090)	3.28*** (11.32)	1.96*** (7.47)	
Panel D: Long-term predictability					
Dependent variable	Observations	PI			
$R_{i,t+120}$	2,088,332	2.43*** (2.42)			
$R_{i,t+1212}$	2,088,332	3.61** (2.19)			
$R_{i,t+12012}$	2,088,332	8.16*** (2.57)			
Panel E: Market state					
Dependent variable	Observations high state	PI high state	Observations low state	PI low state	
$R_{i,t}$	Volatility	863,531	1.07*** (2.80)	779,723	0.32*** (3.94)
$R_{i,t+1}$	Liquidity	1,027,066	1.04*** (4.07)	1,097,253	0.96*** (2.95)
$R_{i,t+12}$	Sentiment	1,046,014	1.39*** (5.35)	969,629	0.75*** (2.48)
Panel F: Nonlinear effect					
Dependent variable	Observations	PI	PI nonlinear		
$R_{i,t}$	Feature-first transformation	2,088,332	0.75*** (4.51)		
$R_{i,t+1}$	Feature-first transformation	2,088,332	0.49*** (2.90)	0.88*** (4.40)	
$R_{i,t+12}$	Peer-first transformation	2,088,332	1.08*** (5.90)		
$R_{i,t+120}$	Peer-first transformation	2,088,332	0.64*** (2.93)	0.59** (2.40)	

Panel A-F report average slopes and Newey-West t -statistics (12 lags) from monthly value-weighted Fama-MacBeth regressions. Monthly returns are expressed in percent. MOM1 and MOM12 follow Jegadeesh (1993) and Jegadeesh and Titman (1993). INDMOM is the industry momentum variable from Moskowitz and Gruber (1999). Lead-Lag is the industry lead-lag signal from Hong et al. (2007). Panel E examines the PI predictability across market states: “low” (“high”) states are months below (above) the cross-sectional median of the previous month’s value of the state variable (VX for volatility, Amihud (2002) for illiquidity, and Baker and Wurgler (2000) for sentiment). Panel F evaluates a nonlinear version of PI constructed from quadratic and cubic transformation of peer-average and peer-deviation characteristics. The sample period is January 1980–December 2024. In the main sample, we exclude “nano” stocks (market equity below the NYSE 1st percentile). The “no microcap” sample excludes firms below the NYSE 20th percentile of market equity. One, two, and three asterisks denote significance at the 10%, 5%, and 1% levels, respectively.

Table 2
Monthly out-of-sample performance.

	NN	GBRT	RF
$R^2_{OOS}^{ML}$	0.322%	0.242%	0.306%
$R^2_{OOS}^{PI}$	0.325%	0.325%	0.325%
$R^2_{OOS}^{AVE}$	0.498%	0.391%	0.427%
DM	2.387	2.153	1.740
p-value	(0.017)	(0.031)	(0.082)

This table reports out-of-sample performance for three machine-learning methods: neural network (NN), gradient-boosted regression trees (GBRT), and random forest (RF). For each method, we report R^2_{OOS} for: (1) a benchmark using 130 firm-specific characteristics (ML); (2) a model using only the Peer Index (PI); and (3) a simple average of the ML and PI forecasts (AVE). The ML models use an expanding-window scheme with an initial 60-month training period (1973–1977) and a 24-month validation period (1978–1979), with the out-of-sample period starting in January 1980. The PI model is estimated with a rolling 84-month window, yielding an out-of-sample start of January 1987. “DM” reports the modified Diebold–Mariano statistic (Gu et al., 2020 adjustment) comparing AVE vs. ML ; Newey–West standard errors are used and p -values appear in parentheses.

8.2 Table 2: Out-of-sample performance vs ML

Object: compare out-of-sample R^2 across (i) ML using only $x_{i,t}$, (ii) PI using $(z_{i,t}, \tilde{x}_{i,t})$, and (iii) the simple average forecast.

- PI alone has slightly higher R^2_{OOS} than each ML benchmark in the table.
- The average forecast (ML+PI) improves R^2_{OOS} further, with statistically significant MSPE improvements (Diebold–Mariano tests).

Interpretation: PI contains predictive information that is complementary to own-characteristic ML forecasts.

8.3 Table 3: Portfolio sorts on PI (equal- and value-weighted)

Object: sort stocks into PI quintiles and form top-minus-bottom portfolios.

- Equal-weighted spreads are very large and remain highly significant after controlling for standard factor models.
- Value-weighted spreads are smaller (consistent with stronger effects in smaller firms) but remain positive and statistically significant.
- “ML-adjusted” PI sorts still deliver significant spreads.

Table 3
Risks and alphas of PI-based portfolios.

	Excess ret.	Alpha	Market	SMB	HML	RMW	CMA	UMD	REV
Panel (A) Equal-weighted PI portfolios (7-factor model)									
Top-minus-bottom	3.33*** (23.03)	3.18*** (23.60)	0.03 (1.01)	0.10** (2.15)	0.06 (1.01)	-0.14** (-2.25)	0.38*** (4.30)	0.25*** (7.75)	-0.18*** (-4.34)
Top-minus-bottom (post 2003)	2.25*** (13.44)	2.24*** (13.81)	-0.05 (-1.17)	0.02 (0.34)	0.09 (1.21)	0.16* (1.80)	0.39*** (5.60)	0.07* (1.70)	-0.06 (-0.98)
Top-minus-bottom (adjust ML)	1.87*** (12.94)	1.79*** (14.00)	0.10*** (3.26)	-0.03 (-0.63)	0.17*** (3.01)	-0.11* (-1.88)	0.25*** (2.98)	0.18*** (6.03)	-0.41*** (-10.43)
Panel (B) Value-weighted PI portfolios (7-factor model)									
Top-minus-bottom	0.89*** (6.05)	0.68*** (5.45)	0.07** (2.47)	-0.08* (-1.78)	0.21*** (3.80)	-0.20*** (-3.59)	0.20** (2.43)	0.41*** (14.25)	-0.18*** (-4.58)
Top-minus-bottom (post 2003)	0.34* (1.73)	0.34** (2.12)	-0.02 (-0.40)	-0.17** (-2.39)	0.30*** (4.20)	0.04 (0.41)	0.30*** (2.81)	0.36*** (8.78)	-0.01 (-0.18)
Top-minus-bottom (adjust ML)	0.53*** (3.76)	0.40*** (3.11)	0.09*** (3.00)	-0.16*** (-3.53)	0.22*** (3.86)	-0.11* (-1.85)	0.14* (1.62)	0.24*** (8.12)	-0.20*** (-6.63)

This table reports the monthly excess returns, factor loadings, and alphas of PI-based portfolios, considering a seven-factor model includes the Fama-French five factors plus the momentum factor and the reversal factor. Panel A and B show the results for the equal-weighted and value-weighted portfolios, respectively. The sample period is Jan 1980-Dec 2024. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 4
Annual alphas of PI-portfolios over longer holding periods.

	Holding period		
	3	6	12
VW top quintile	0.36 (0.49)	0.85 (1.49)	0.79* (1.73)
VW bottom quintile	-4.01*** (-4.84)	-2.47*** (-3.68)	-1.44*** (-2.82)
VW spread portfolios	4.37*** (7.46)	3.31*** (8.77)	2.23*** (8.67)
VW spread portfolios (post 2003)	2.12*** (2.84)	1.63*** (3.15)	1.28*** (3.84)

This table reports the annual alphas (in percentage) and their t-values (in parentheses) obtained from regressing monthly value-weighted PI portfolio returns on Fama-French five factors plus the momentum factor and the reversal factor. Annual alphas are obtained by multiplying monthly alphas by 12. We follow the technique from [Jegadeesh and Titman \(1993\)](#) to deal with holding periods longer than one month. The sample period is Jan 1980 to Dec 2024. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

Interpretation: the PI signal is economically meaningful in trading-portfolio space and not simply an artifact of known factor exposures.

8.4 Figure 1: Cumulative performance of PI portfolios

Object: cumulative returns of the PI strategy relative to the market.

Interpretation: the PI-based long-short portfolio delivers persistent cumulative gains, indicating that the predictive content is stable enough to accumulate economically meaningful performance.

8.5 Table 4: Longer holding periods

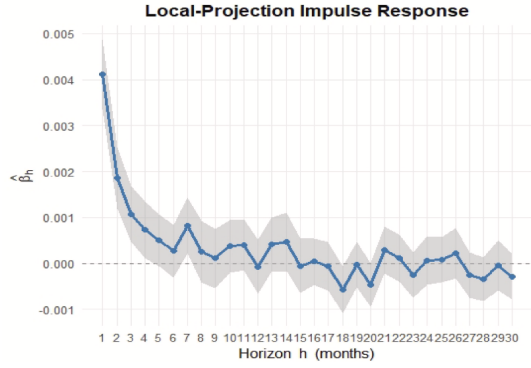
Object: annualized alphas for holding PI portfolios over 3-, 6-, and 12-month horizons.

Interpretation: the spread alphas remain strongly positive even for longer holding periods, which is consistent with *slow diffusion* (not quick reversal).

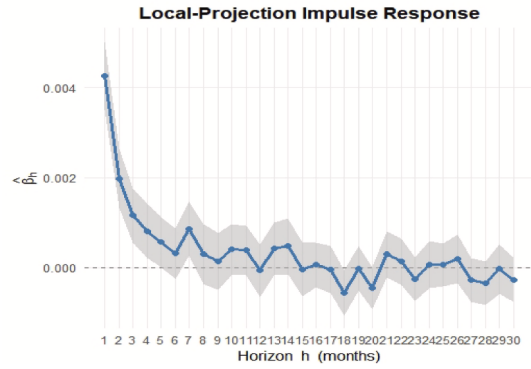
8.6 Figure 2: Model complexity vs realized OOS performance

Object: how portfolio return/Sharpe and model R^2_{OOS} vary with model complexity $c = P/T$.

Interpretation: performance is not monotone in complexity; disciplined regularization matters. The figure supports that the PI construction balances fit and stability in a way that performs well out of sample.



(a) Panel A: Impulse response of $PI_{i,t}$ on future returns $R_{i,t+h}$ with a 95% confidence interval for horizons $h = 1$ to 30 months. PI is scaled by 10.



(b) Panel B: Impulse response of $PI_{i,t}$ shocks on future returns $R_{i,t+h}$ with a 95% confidence interval for horizons $h = 1$ to 30 months. The shock variable is scaled by 10.

Fig. 3. Lag-Augmented Local Projection (LA-LP).

8.11 Table 8: where PI is strongest (information environment heterogeneity)

Object: interaction regressions showing whether PI is stronger for: smaller firms, fewer analysts, higher forecast dispersion, less news coverage, and smaller changes in news coverage.

Interpretation: PI is stronger exactly when attention and information processing are plausibly weaker, supporting an underreaction channel.

Table 7
Predicting firm-level earnings surprises.

	(1) One-quarter-ahead	(2) Four-quarter-ahead	(3) Eight-quarter-ahead
PI	0.026*** (21.29)	0.0112*** (9.38)	0.001*** (2.90)
ML	0.022*** (16.74)	-0.009*** (-8.00)	-0.008*** (-6.55)
Observations	639,525	628,144	556,180
Adj. R ²	0.032	0.028	0.023
Controls	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes

This table reports the results of panel regression for predicting multiple-step-ahead earnings surprises. The earnings surprise is measured by the standardized unexpected earnings (SUE), which is computed as the I/B/E/S actual earnings minus the median forecasted earnings divided by fiscal-quarter-end market cap. If earnings forecast is not available, then the unexpected earnings is computed as the I/B/E/S actual earnings minus the seasonally differenced quarterly earnings. Control variables include the firm's current SUE, and its industry-average counterpart, along with size and the book-to-market ratio. Both regressions incorporate individual and quarter-fixed effects. All explanatory variables are cross-sectionally standardized. For brevity, the coefficients on these controls are not reported. *t*-statistics based on standard errors clustered by entity are shown below coefficient estimates. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 8
Cross-sectional variation in PI effect.

Dependent variable: One-month-ahead return	(1)	(2)	(3)	(4)	(5)	(6)
PI	0.93*** (3.68)	2.46*** (10.04)	2.87*** (8.11)	2.39*** (7.97)	1.52*** (5.44)	1.64*** (5.73)
PI*Dummy(Size < 30%)	2.23*** (4.10)	2.13*** (10.95)		1.20*** (3.73)		
PI*Dummy(Analyst < 30%)	0.10 (0.22)				0.82*** (4.98)	
PI*Dummy(Dispersion > 70%)	0.28 (1.10)					1.28*** (5.97)
PI*Dummy(News < 30%)	0.92*** (3.16)					0.83*** (3.84)
PI*Dummy(News < 30%)	0.42* (1.69)					0.83*** (3.84)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Average# of stocks	3995	3963	3963	3963	3995	3995
Average adjusted R ²	0.051	0.036	0.034	0.042	0.040	0.040

This table reports regression results of the heterogeneity tests. We use five variables to proxy the speed of information flow, including size, number of analysts, analyst dispersion, news mentions and change in news mentions. The two news-based attention variables are available since 2003. For size, number of analysts, news, and Δ news, we create an indicator variable equal to 1 if the stock is below the bottom 30%, and 0 otherwise. For analyst dispersion, we create an indicator variable equal to 1 if the stock is above the top 30%, and 0 otherwise. In columns 2-6, we conduct FM regression by regressing future returns on PI and one of the interaction terms. In column 1, we conduct FM regression by regressing future returns on PI and all five interaction terms. Control variables include past one-month returns, past 12-month returns (excluding the most recent month), size and book-to-market. FM coefficients are the means of the monthly estimated coefficients*100; *t*-statistics employ Newey-West adjustments of 12 lags.

Table 6
Predicting industry-level fundamentals & Returns.

	$SUE_{ind,j,t+1}$	$R_{ind,j,t+1}$
PI	0.004*** (22.27)	0.920*** (8.61)
ML	0.001*** (5.77)	0.165 (1.52)
Controls	Yes	Yes
Fixed effect	Yes	Yes
Adjusted R ²	0.067	0.112
Observations	639,525	2,086,995

This table reports the results of panel regression for predicting one-step-ahead earnings surprises and excess returns at the industry level. The earnings surprise is measured by the standardized unexpected earnings (SUE), which is computed as the I/B/E/S actual earnings minus median forecasted earnings divided by fiscal-quarter-end market cap. If earnings forecast is not available, then the unexpected earnings is computed as the I/B/E/S actual earnings minus the seasonally differenced quarterly earnings. In the regression for predicting one-step-ahead earnings surprises, we include both time and entity fixed effects. Control variables include current industry average SUE, together with industry average size and book-to-market. *t*-statistics based on standard errors clustered by entity are shown below coefficient estimates. The second column of the table shows the average slopes*100; *t*-statistics with Newey-West adjustments of 12 lags from monthly Fama-MacBeth regressions for industry return. Control variables include lagged industry-average returns, industry-average size, and the book-to-market ratio. All explanatory variables are cross-sectionally standardized. The adjusted R² is the average across all cross-sectional regressions. For brevity, the coefficients on these controls are not reported. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 9
Cross firm momentum effects and Fama-MacBeth cross-sectional regressions.

Dependent variable	Observations	PI	INDMOM	Lead-lag	SA_Ret	Cus_Ret
$R_{i,t+1}$	2,088,332		0.21 (1.40)			
$R_{i,t+1}$	2,088,332	0.98*** (6.53)				
$R_{i,t+1}$	2,088,332			0.56** (3.52)		
$R_{i,t+1}$	2,088,332	0.86*** (6.45)		0.31* (1.72)		
$R_{i,t+1}$	1,674,481				1.03*** (4.62)	
$R_{i,t+1}$	1,674,481	0.84*** (7.20)			0.86*** (3.81)	
$R_{i,t+1}$	2,000,816					0.39*** (2.90)
$R_{i,t+1}$	2,000,816	0.72*** (5.50)				0.36*** (2.60)
$R_{i,t+1}$	1,674,481	0.87*** (5.92)	-0.57* (-1.88)	0.13 (0.46)	0.90*** (3.89)	0.24 (1.61)

The table shows the average slopes*100; *t*-statistics with Newey-West adjustments of 12 lags from monthly value-weighted Fama-MacBeth regressions. Control variables include size, book-to-market ratio, short-term reversal, and momentum. The sample period is January 1980-Dec 2024. For the main sample, we exclude "nano" stocks whose market capitalization is below the 1st percentile of the NYSE sample. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

8.12 Table 9: relation to cross-firm momentum spillovers

Object: compare PI to known cross-firm signals: industry momentum, lead-lag via shared analysts, and customer momentum.

Interpretation: PI remains robust when controlling for these signals and partly subsumes some of them. Thus PI is not merely a repackaging of previously proposed cross-stock momentum spillovers.

9 Asymmetric networks and the symmetric–antisymmetric decomposition

9.1 Decomposing a directed network

For an asymmetric (directed) network with position matrix L , define

$$L_{\text{sym}} = \frac{1}{2}(L + L'), \quad L_{\text{asym}} = \frac{1}{2}(L - L'), \quad L = L_{\text{sym}} + L_{\text{asym}}. \quad (18)$$

Applying L to a signal vector $x_t \in \mathbb{R}^N$ yields two peer-average legs:

$$z_{\text{sym}} = L_{\text{sym}}x_t, \quad z_{\text{asym}} = L_{\text{asym}}x_t. \quad (19)$$

The first corresponds to symmetric peer exposure (a beta-style leg); the second captures cross-directional timing/alpha embedded in the directed part of the network.

For peer-adjusted signals, let $M = I - L$. Then

$$M_{\text{sym}} = \frac{1}{2}(M + M') = I - L_{\text{sym}}, \quad M_{\text{asym}} = \frac{1}{2}(M - M') = -L_{\text{asym}}. \quad (20)$$

Hence the peer-adjusted strategy decomposes as

$$(I - L)x_t = (I - L_{\text{sym}})x_t + (-L_{\text{asym}})x_t. \quad (21)$$

9.2 Table 10: economic meaning of the four components

Table 10 summarizes the interpretation:

- $L_{\text{sym}}x_t$: symmetric peer average $\Rightarrow$ characteristic-implied factor beta.
- $L_{\text{asym}}x_t$: antisymmetric peer average $\Rightarrow$ factor-neutral, cross-directional timing/alpha.
- $(I - L_{\text{sym}})x_t$: symmetric peer deviation $\Rightarrow$ beta to peer-adjusted characteristics.
- $(I - L_{\text{asym}})x_t$: mixture $\Rightarrow$ own-signal beta plus alpha from peer-adjusted strategy.

9.3 Table 11: drop-one-out importance

Table 11 drops one component at a time and re-computes OOS R^2 and portfolio returns.

- In asymmetric networks (shared analysts; customer–supplier), the antisymmetric peer-average component is especially valuable: dropping it hurts the most.

- Dropping the antisymmetric deviation can even *improve* performance, suggesting it may add noise in some settings.

10 Appendix tables (12–16): what they add

- **Table 12:** the complete list of characteristics and their categorizations. Key message: PI is built from a *broad* set of signals, not a narrow anomaly.
- **Table 13:** descriptive statistics of characteristics and their correlations with PI. Key message: correlations are generally modest; PI is not trivially one characteristic in disguise.
- **Table 14:** descriptive statistics for peer-average and deviation variables used in PI construction and average ridge coefficients. Key message: both peer averages and deviations receive non-trivial weights, consistent with the “two-channel” structure.
- **Table 15:** PI abnormal returns are not explained away by simple ETF exposures to Quality/Value/Momentum; alpha remains.
- **Table 16:** sentiment variables do not subsume PI; PI remains significant.

11 My consolidated takeaways (what I would remember)

1. **Conceptual contribution:** expected returns can depend on *peer-transformed* characteristics, not only on own characteristics. This yields two interpretable channels: peer strength and relative position.
2. **Empirical contribution:** PI predicts returns and earnings surprises at short and long horizons, is robust to many controls, and is not subsumed by own-characteristic ML forecasts.
3. **Mechanism evidence:** dynamics and heterogeneity strongly point to *underreaction* / *slow information diffusion* in peer information.
4. **Network perspective:** directed networks naturally create a symmetric–antisymmetric decomposition, and the antisymmetric (directional) peer-average component can be a distinct and powerful source of predictability.

12 Conclusion and intuition (Avramov–Ge–Li–Linton, JFE 2026)

12.1 Coherent summary: what the paper ultimately establishes

The paper’s central conclusion is that a large and economically meaningful component of cross-sectional return predictability is *network-structured*: information from *economically linked peers* helps forecast a firm’s future returns and fundamentals, and this information is not fully captured by models that rely only on a firm’s own characteristics.

The authors operationalize this insight by constructing a **Peer Index (PI)** that maps peer information into a stock’s expected return through two interpretable channels:

$$z_{i,t} \equiv (\text{peer-average characteristics: “peer strength”}), \quad (2)$$

$$\tilde{x}_{i,t} \equiv x_{i,t} - z_{i,t} \quad (\text{deviation from peers: “relative position”}), \quad (3)$$

$$PI_{i,t} \equiv z'_{i,t}\hat{\beta}_z + \tilde{x}'_{i,t}\hat{\beta}_x \quad (\text{estimated via ridge}). \quad (4)$$

This dual-channel construction matters because it clarifies *why* one firm’s characteristics can predict another firm’s return: firms inherit information from their peer groups, and being unusually positioned within a peer group is informative on top of the group mean.

12.2 Key empirical conclusions (what to remember)

1. **PI predicts returns strongly and robustly.** Portfolio sorts on PI generate large and statistically strong spreads. For example (Table 3), the monthly top-minus-bottom PI portfolio earns about

$$\text{EW: } 3.33\% \text{ (t=23.03)}, \quad \text{VW: } 0.89\% \text{ (t=6.05)},$$

with large factor-model alphas (seven-factor model):

$$\alpha_{\text{EW}} \approx 3.18\% \text{ (t=23.60)}, \quad \alpha_{\text{VW}} \approx 0.68\% \text{ (t=5.45)}.$$

Even after removing the component of PI that is predictable by an own-characteristics ML forecast, the remaining “ML-adjusted PI” still delivers strong spreads (Table 3).

2. **PI adds information beyond powerful ML predictors built from firm-level characteristics.** Out-of-sample (Table 2), PI alone achieves $R^2_{\text{OOS}} \approx 0.325\%$ and combining PI with ML forecasts improves forecasting accuracy further:

$$R^2_{\text{OOS}}(\text{ML+PI}) \in [0.391\%, 0.498\%]$$

depending on the ML benchmark. This supports the interpretation that PI captures *genuine cross-stock information* that does not reduce to a better nonlinear transformation of $x_{i,t}$ alone.

3. **PI forecasts fundamentals (earnings surprises), not only returns.** PI predicts industry-level earnings surprises and industry returns (Table 6) and also predicts firm-level earnings surprises at one-, four-, and eight-quarter horizons (Table 7), with PI remaining positive at longer horizons even when ML signals weaken or flip sign. This links the return predictability to slow-moving information about cash flows.
4. **The dynamics support underreaction / slow diffusion rather than reversal.** Lag-augmented local projections (Fig. 3) show that positive PI innovations are followed by higher future returns that *gradually decay* toward zero *without a reversal*. This time-series pattern is consistent with sluggish incorporation of peer information into prices.
5. **The effect is stronger in weak information environments (mechanism evidence).**

Heterogeneity tests (Table 8) show larger PI effects when information processing is plausibly harder or attention is weaker: smaller firms, fewer analysts, higher analyst dispersion, fewer news mentions, and smaller changes in news coverage.

6. **The effect is not easily explained by standard risk compensation or sentiment.**

The paper presents spanning and robustness evidence indicating that PI’s predictive content is not simply a repackaging of standard factor exposures, and sentiment-based controls do not eliminate the PI effect.

12.3 Intuition: why PI works

A coherent economic interpretation is **limited attention to peer-relevant information**. Investors (and, to some extent, analysts) overweight firm-specific signals and underweight information embedded in economically linked peers. Because peer information is diffuse (it requires aggregation across linked firms and interpretation of relative positioning), it enters prices gradually.

The PI construction is therefore powerful because it explicitly encodes the two pieces of information that are naturally slow to process:

- **Peer strength (average channel):** if the peer group exhibits strong profitability, investment opportunities, or other value-relevant characteristics, a focal firm is partly exposed to that state through shared economic conditions and linkages.
- **Relative position (deviation channel):** if a firm is unusually cheap, unusually constrained, or unusually profitable relative to peers, this contains incremental information (mispricing or future fundamentals) beyond the peer-group average.

Because markets do not instantly process these network-based signals, PI predicts both *future earnings surprises* and *future returns*, and the absence of long-horizon reversal aligns with underreaction rather than transient overreaction.

12.4 Directed networks and “alpha” vs “beta” (how to interpret the asymmetric-network evidence)

Using symmetric industry peers, the peer linkage matrix is symmetric, so the directional component is mechanically absent. When the peer network is genuinely directed (customer–supplier links or shared-analyst co-coverage), the symmetric–antisymmetric decomposition clarifies that part of PI can be interpreted as:

- a **factor-exposure (“beta”)** component tied to symmetric peer averages, and
- a **factor-neutral directional (“alpha”)** component tied to the antisymmetric part of the network.

Drop-one-out results (Table 11) indicate that, in asymmetric networks, the directional peer-average component can be especially important, highlighting that economically directed links can generate predictability beyond symmetric industry clustering.

12.5 Takeaway

Cross-stock predictability is network-based: an economically structured peer index that combines peer strength and within-peer relative position forecasts returns and earnings, improves out-of-sample performance beyond own-stock ML predictors, and exhibits slow-decaying (non-reversing) dynamics consistent with underreaction to peer information.